RFID-Integrated Child safety & attendance system

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Abstract-

Child safety during school transport and outdoor activities is a matter of utmost concern. Traditional attendance systems fall short in providing real-time monitoring, live geolocation tracking, and instant alerts in emergencies or when children breach predefined zones. To address these limitations, this paper proposes a smart IoT-based system that integrates automated attendance with continuous location tracking using affordable components and cloud connectivity. The system comprises three primary modules: a Scanning Bench, where children scan RFID tags verified by an ESP32 microcontroller; a Live Tracker, equipped with an ESP32, GPS module, and power supply to capture and transmit location data; and a Display Setup that receives GPS coordinates via the ESP-NOW protocol and displays real-time positions on a Google Maps-integrated web dashboard. This setup enables fast, wireless communication, centralized monitoring, and geofencing alerts without relying on mobile phones. The RFID system is linked to a cloud-based database to manage child identities at scale, offering a robust and scalable solution. Furthermore, the architecture supports future upgrades with GSM modules for long-range communication. By combining IoT, RFID, and GPS technologies, this system provides a comprehensive, low-cost solution for enhancing child safety and ensuring efficient monitoring during school commutes and outdoor activities.

Keyword: RFID, GPS, ESP32, ESP-NOW, Firebase, IoT, child safety, attendance system

INTRODUCTION

The safety of school-going children, particularly during transportation or field activities, is a growing concern for educational institutions and parents alike. Traditional attendance systems, which rely on manual entry or passive RFID scanning, provide little to no visibility into a child's real-time location. This becomes critically problematic in cases where a child goes missing, deviates from a designated area, or fails to reach school on time. Recent technological advancements in the Internet of Things (IoT) have led to several attempts at integrating location tracking and automated attendance solutions. For instance, Sharma et al. [1] proposed an RFID-based smart attendance system for schools that automated entry logs but lacked live location feedback or alert mechanisms. In another study, Gupta et al. [2] presented a GSM-GPS-based child tracking system capable of sending the location via SMS, although the system was limited to single-unit monitoring and lacked a centralized visualization platform.

More comprehensive models, such as that by Mehta [3], incorporated GPS-based tracking within school buses and allowed parents to receive SMS alerts upon arrival or departure. However, this model tracked the vehicle, not the individual child, which fails to resolve scenarios where a student accidentally or intentionally exits the vehicle. Similarly, projects like the one by Joshi et al. [5] integrated RFID with cloud storage, enabling attendance records to be synced online but still did not address live child monitoring.

There have also been studies exploring the use of Wi-Fi or Bluetooth for local communication. Rathi et al. [4] demonstrated the use of the **ESP-NOW protocol**, an efficient low-power, peer-to-peer wireless communication method for ESP32 microcontrollers. This protocol allows devices to transmit small payloads without requiring a router or internet connection—making it a suitable choice for short-range GPS transmission in child safety systems.

Despite these efforts, a significant gap remains: a need for a cost-effective, scalable, and portable system that not only marks a child's attendance but also continuously monitors their location, visually displays it in real-time, and raises alerts when safety boundaries are breached.

To address these gaps, we propose an RFID-integrated child safety and attendance system using ESP32 microcontrollers. It combines GPS tracking, ESP-NOW wireless data transfer, and Google Maps visualization to ensure real-time child monitoring without the dependency on mobile phones or SIM-based GSM networks.

The proposed RFID-integrated child safety and attendance system is built upon a combination of carefully selected hardware and software components that together ensure seamless functionality. At the core of the hardware setup are ESP32 microcontroller modules, which are utilized both

at the scanning bench and embedded in student ID cards for wireless data transmission. The RC522 RFID module serves to read the unique identifiers associated with each child's ID card during entry, initiating the attendance logging process. To achieve accurate real-time geolocation, a NEO-6M GPS module is employed, continuously capturing and updating the child's position. This entire system is powered by a portable energy solution comprising a 2500 mAh Li-Ion battery, managed through a TP4056 charging module and a MT3608 voltage booster to ensure consistent power delivery throughout the day. Visual and auditory cues, such as LEDs and buzzers, are incorporated to provide instant feedback upon successful RFID scans or alert scenarios.

On the software side, the architecture relies on the Arduino IDE for the development and deployment of firmware across the ESP32 devices. Communication between modules is handled via ESP-NOW, a lightweight and efficient peer-to-peer wireless protocol ideal for short-range data transmission without the need for an active Wi-Fi network. The Firebase platform is integrated as a cloud-based, real-time database to store and sync attendance records and GPS data, facilitating instantaneous access and secure backups. Lastly, a web-based dashboard is implemented to visualize both attendance logs and live GPS locations, offering school administrators and parents a user-friendly interface to monitor each child's safety status effectively.

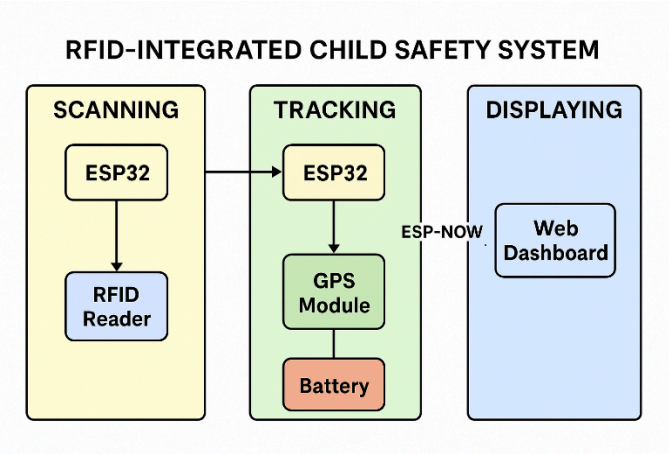

Figure 1: Shows how the system works together

The methodology of the RFID-integrated child safety and attendance system follows a streamlined sequence that begins the moment a student interacts with the scanning infrastructure. The process initiates when a child scans their RFID-enabled ID card at the designated scanning bench, prompting the ESP32 microcontroller to read the RFID data and fetch the associated student details. Upon successful verification, the system logs the attendance entry while simultaneously activating the GPS tracking module. The GPS module, based on the NEO-6M chipset, continuously collects the child's geographical coordinates at regular intervals. These coordinates are transmitted in real-time using the ESP-NOW communication protocol, which enables fast, local wireless data exchange without the need for an internet connection or router.
As data flows through the system, it reaches the Firebase cloud backend where it is securely stored and synchronized across devices. The use of Firebase ensures that the system maintains a consistent and accessible database for both real-time operations and archival purposes. On the frontend, a custom-built web interface receives the live updates and visually represents the child's location on a map, offering intuitive access for parents and school staff. The integration of these components results in a robust system where attendance is automatically logged, child location is continually tracked, and updates are delivered instantaneously to authorized viewers, all while maintaining low power consumption and high communication efficiency.

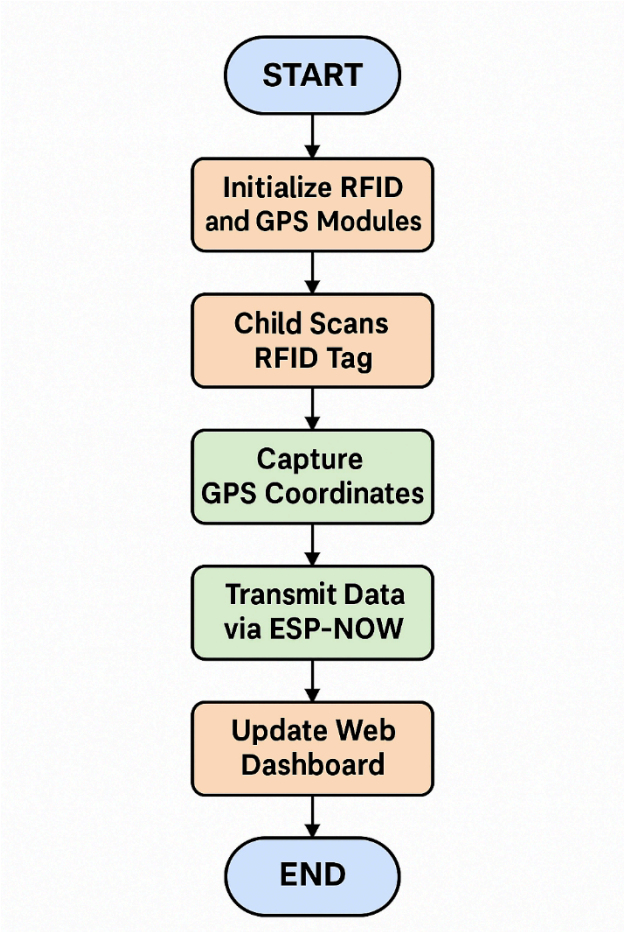

Figure 2: Explains the flow of the system

The implementation of the RFID-integrated child safety and attendance system has yielded promising results that validate its feasibility and effectiveness in real-world educational settings. The developed prototype reliably captures and transmits live GPS updates from the child's tracker, effectively reflecting the child's real-time location on an integrated Google Maps interface. This functionality allows administrators and parents to visualize the movement of children with a high degree of spatial accuracy. The system also successfully employs RFID technology for identity mapping, using unique RFID UIDs to maintain a scalable and organized database of student entries. Leveraging the ESP-NOW protocol, the system ensures rapid, battery-efficient wireless data transfer, maintaining minimal latency-typically around 30 milliseconds in local conditions-which supports real-time monitoring without significant delay.

Firebase integration further strengthens the system by providing real-time data synchronization and secure cloud storage for attendance and location logs. Initial testing confirmed that the GPS module delivers location accuracy

within approximately two meters under open-sky conditions, with reliable position updates every few seconds. However, it was observed that in denser urban environments or indoors, the precision of GPS readings can decrease slightly, which aligns with known limitations of satellite-based positioning systems.

In terms of advantages, the system is notably affordable and highly scalable, making it an attractive solution for wide-scale deployment across schools with limited budgets. Its design avoids reliance on continuous internet connectivity at the tracker level, significantly reducing operational costs and simplifying deployment logistics. Moreover, the hardware and software components are easy to integrate into existing school infrastructure, which further enhances the system's practical value.

Implementation of Scanning Bench: (ESP32 – RC522 – Buzzer – RFID Tags)

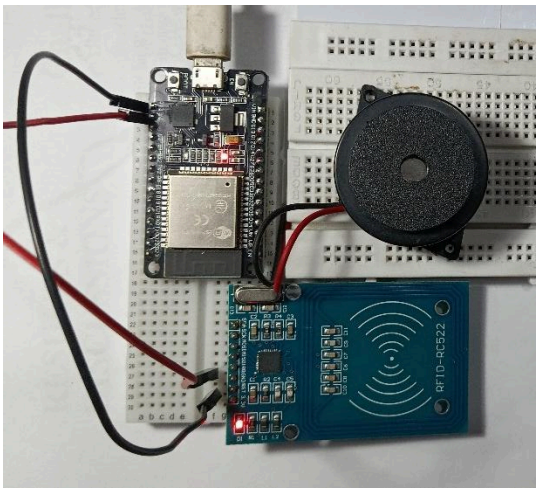

Figure 3.1: Scanning Bench

Implementation of GPS Tracker:
(ESP32 – NEO-6M – Battery)

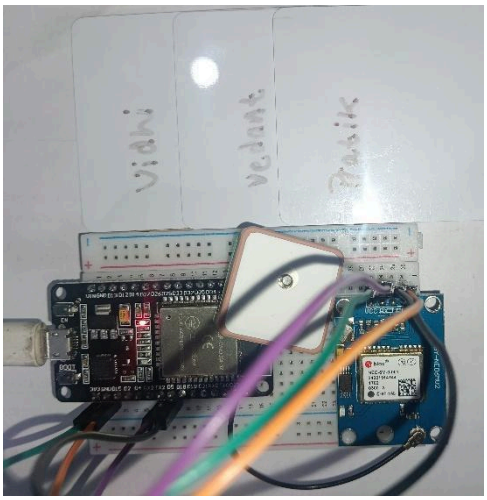

Figure 3.2: GPS Tracker

Overall Setup Implementation:
(ESP32, Arduino Uno (For temporary implementation of Scanning Bench), RC522 (RFID Module), RFID Cards, Neo-6M (GPS Module), Buzzer, Breadboard, Battery Module for GPS (Not included yet), Jumping Wires

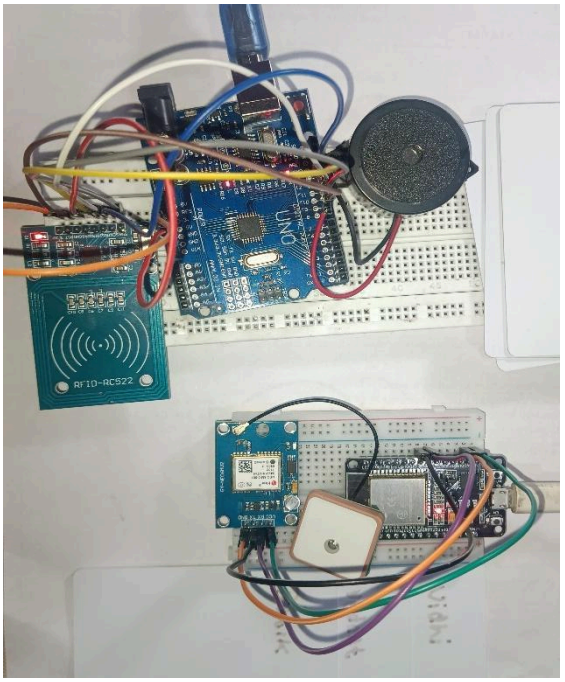

Figure 3.3: Combined System

Web Dashboard displaying Live Location on Google Maps:

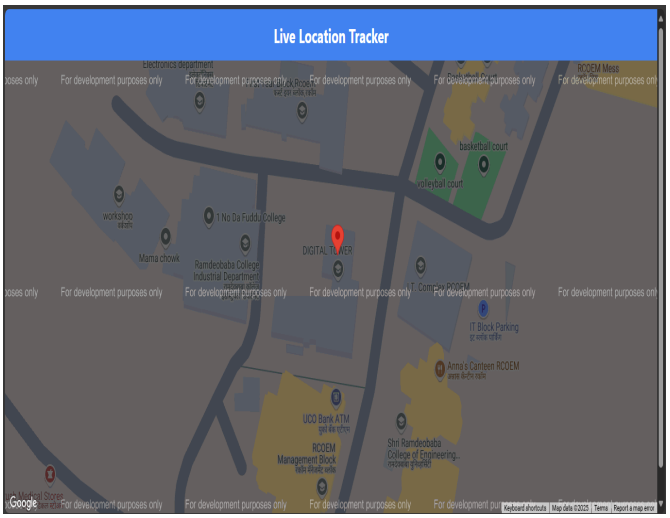

Figure 3.4: Web Dashboard

Despite these strengths, some challenges remain. One critical area for improvement is battery life, as the GPS module's continuous operation demands optimized power management to support full-day tracking without recharge interruptions. Additionally, while ESP-NOW offers a robust communication channel, the system's performance may degrade in environments with substantial obstructions, such as tunnels or thick concrete structures. Lastly, the ongoing need for enhanced data security measures is evident.

As the system handles sensitive information regarding children's identities and locations, implementing stronger encryption and access control mechanisms is essential to safeguard against unauthorized access or data breaches.

CONCLUSIONS

The RFID-Integrated Child Safety and Attendance System, as demonstrated through this project, underscores the practical viability and operational effectiveness of utilizing low-cost Internet of Things (IoT) components for real-time child monitoring within educational settings. By synergizing RFID-based identity verification, continuous GPS tracking, and ESP-NOW wireless communication, the system offers a responsive and scalable solution that meets the critical needs of both safety and administrative management. Real-time updates on a child's location, visualized through a live web dashboard integrated with Google Maps, provide a transparent and immediate method for tracking movements, thereby eliminating reliance on manual attendance logs and significantly reducing the risk associated with location ambiguity during school hours or transit. The system not only enhances administrative accuracy but also strengthens the sense of security for parents and school officials alike. Its cost-efficiency and minimal infrastructural demands make it an attractive candidate for broader adoption across a variety of institutional settings. Looking forward, further refinements in battery optimization and data security will augment its robustness, paving the way for more comprehensive, tech-enabled safety ecosystems in education.

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